

Full Stack Internship Assignment

Project: Smart Leads Dashboard

Objective

Build a full-stack Lead Management Dashboard using the MERN stack with clean architecture, scalable code practices, and a professional user experience.

The purpose of this assignment is to evaluate:

- Frontend development skills
 - Backend/API design
 - Database modeling
 - State management
 - Code quality
 - Problem-solving ability
 - Real-world engineering practices
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Technical Requirements

Mandatory Tech Stack

Frontend

- React.js
- TypeScript (**mandatory**)
- TailwindCSS

Backend

- Node.js
 - Express.js
 - TypeScript (**mandatory**)
 - MongoDB + Mongoose
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Important Note

- Plain JavaScript submissions will be **rejected**
- Proper TypeScript usage is mandatory
- Interfaces/types should be properly defined
- Usage of **any** should be minimized and justified where necessary

Core Features

1. Authentication System

Implement JWT-based authentication.

Features

- User Registration
- User Login
- Protected Routes
- Password Hashing using bcrypt
- Auth Middleware

Expectations

- Secure authentication flow
 - Proper token handling
 - Clean middleware structure
 - Validation and error handling
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2. Leads Management (CRUD)

Create a Lead Management module.

Lead Fields

- Name
- Email
- Status
 - New
 - Contacted
 - Qualified
 - Lost
- Source
 - Website
 - Instagram
 - Referral
- Created At

Functionalities

- Create Lead
- Update Lead
- Delete Lead

- View Leads List
 - View Single Lead Details
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3. Advanced Filtering & Search

This is one of the most important parts of the assignment.

Required Filters

- Filter by Status
- Filter by Source
- Search by Name or Email
- Sort by:
 - Latest
 - Oldest

Important

Multiple filters should work together.

Example

- Status = Qualified
 - Source = Instagram
 - Search = "Rahul"
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4. Pagination

Requirements

- Backend pagination is mandatory
- Limit: 10 records per page

Expected Implementation

- Proper use of `skip` and `limit`
 - Pagination metadata in API response
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5. Frontend UI Requirements

Build a clean and responsive dashboard UI.

Mandatory UI Expectations

- Responsive Design
 - Reusable Components
 - Proper Folder Structure
 - Loading States
 - Empty States
 - Error Handling UI
 - Form Validation
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6. API Standards

Requirements

- RESTful API structure
 - Proper status codes
 - Centralized error handling
 - Request validation
 - Clean response format
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Mandatory Additional Features

The following features are mandatory and will not be considered bonus features:

- Debounced Search
 - CSV Export Functionality
 - Role-Based Access Control
 - Admin
 - Sales User
 - Docker Setup
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Bonus Feature

- Dark Mode Support
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Submission Requirements

Candidates must submit:

- GitHub Repository URL
- Updated Resume
- Proper README.md

- [.env.example](#)
- [API Documentation](#)
- [Setup Instructions](#)
- [Deployment Link \(Preferred\)](#)

Submission Email

Please send your submission to:

ritik.yadav@servicehive.tech

Subject Format:

MERN Internship Assignment Submission - Your Name

Evaluation Criteria

Candidates will primarily be evaluated on:

- Code Quality
 - Project Structure
 - TypeScript Usage
 - Reusability
 - API Design
 - UI/UX Quality
 - Error Handling
 - Scalability
 - Git Commit Quality
 - Problem Solving Approach
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Automatic Rejection Criteria

Submissions may be rejected if they contain:

- Plain JavaScript instead of TypeScript
 - Poor folder structure
 - No validation
 - Hardcoded values/URLs
 - No loading/error states
 - Extremely large components
 - Copied/template-style code
 - Improper TypeScript usage
 - Missing interfaces/types
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